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OptiProductio

Arduino-Based Real-Time WIP Monitoring and Bottleneck Detection

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Team Details

Batch number: 2 **Domain:** Industrial Engineering & Management

Project: OptiProductio: Arduino-Based Real-Time WIP Monitoring and Bottleneck Detection

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Objectives

- **Main objective:** build a low-cost prototype on a 3-station model line (S1, S2, S3: Station 1, 2, 3) that measures, in real time, production count (parts finished), cycle time (time a station takes per part), inventory (raw material left in the bin) and WIP (work in progress: parts inside the line, not yet finished). Use these to show which of four bottlenecks (the slowest point that holds up the whole line) is causing a delay: a slow station, a breakdown, an empty raw-material bin or a burst of rejects (bad parts). Then test a WIP limit (a cap on parts allowed inside the line) that stops parts piling up.

PO (project objective)	Specific objective	Measurable target	By
PO1	Count parts at all 3 stations with IR sensors	Count error ≤ 2% against a manual tally, over three 10-minute runs	Phase II
PO2	Live dashboard: production count, cycle time per station, WIP, bin level, downtime (time the line is stopped), OEE proxy (approximate overall efficiency)	Refreshes every 2 s or faster; all 6 values on one screen	Phase II
PO3	Identify the bottleneck, then rebalance the line (move work from the slowest station so all stations take similar time)	Correct cause shown in 4 of 4 staged tests (each bottleneck created on purpose: slow S2, breakdown, empty bin, rejects); throughput (good parts per minute) and flow time (time for one part to cross the line) compared before and after, 3 runs each	Phase III
PO4	WIP-limit alert on the red LED, buzzer and dashboard	Alert triggers every time WIP reaches the limit; results compared for limits of 2, 4 and 6 parts	Phase III

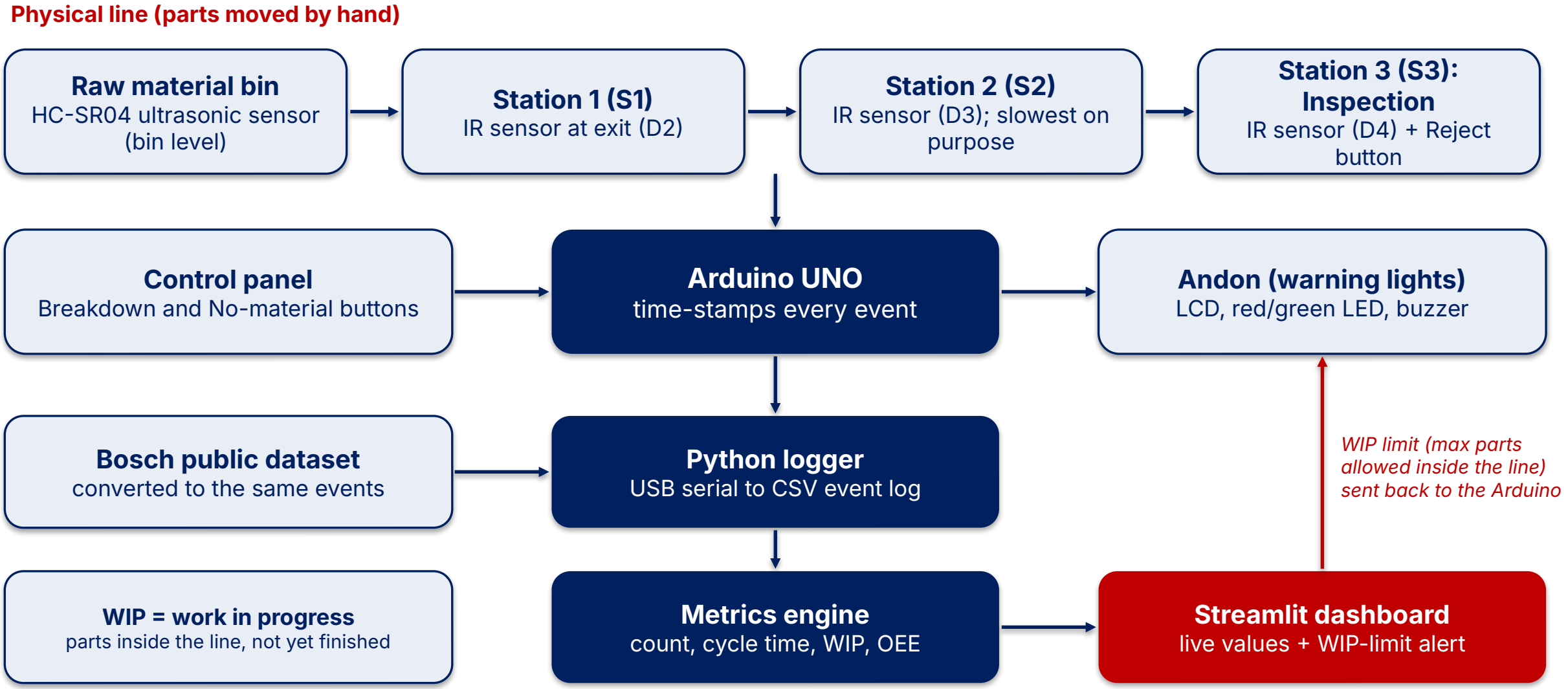
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Methodology

- **Design:** experimental prototype. A model line of 3 stations (S1, S2, S3) in a row, with small identical parts moved by hand. The task at S2 is made longer on purpose so that S2 becomes the bottleneck (the slowest step, where parts pile up). Each run lasts 10 minutes and is repeated 3 times: baseline, then rebalanced (part of S2's work moved to S1 so stations take similar time), then with a WIP limit of 2, 4 and 6 parts.
- **Data collection:** Arduino UNO with 3 FC-51 IR sensors at the station exits (time-stamp each part leaving), an HC-SR04 ultrasonic sensor above the raw-material bin (parts left) and 3 push buttons for Breakdown, No material and Reject. Events go over USB serial (9600 baud) to a Python logger that writes a CSV event log. An LCD, red/green LEDs and a buzzer act as the andon (signal lights that warn everyone on the line).
- **Data analysis:** from the event log we compute production count, cycle time (time a station actually works on one part, not counting time spent waiting), the bottleneck (station with the longest cycle time), WIP, throughput (good parts per minute) and flow time (time one part takes to cross the whole line). Cross-checks: Little's law [2] (average WIP = throughput × flow time) and an OEE proxy [1] (overall equipment effectiveness, approximate: availability × performance × quality).
- **Validation:** counts within 2% of a manual tally; the dashboard must name the right cause in 4 staged tests (each bottleneck created on purpose: slow S2, breakdown, empty bin, burst of rejects); the same code is run on the public Bosch production line dataset (Kaggle; first 50,000 rows, 3 busiest stations of line L3) [3], [4].
- **Tools:** Arduino IDE (C++); Python with pyserial, pandas and Streamlit (dashboard refreshed every 2 s); Excel for result tables.

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Methodology Flowchart



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References

[1] Ng Corrales, L. del C., Lambán, M. P., Hernandez Korner, M. E., & Royo, J. (2020). Overall equipment effectiveness: Systematic literature review and overview of different approaches. Applied Sciences, 10(18), 6469. https://doi.org/10.3390/app10186469

[2] Little, J. D. C. (2011). Little's law as viewed on its 50th anniversary. Operations Research, 59(3), 536–549. https://doi.org/10.1287/opre.1110.0940

[3] Mangal, A., & Kumar, N. (2017). Using big data to enhance the Bosch production line performance: A Kaggle challenge. arXiv. https://arxiv.org/abs/1701.00705

[4] Bosch. (2016). Bosch production line performance [Data set]. Kaggle. https://www.kaggle.com/c/bosch-production-line-performance

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